

A 1:10 Autonomous Car for FEB Autonomous

Components, options and budget

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The ask

One 1:10-scale autonomous car on the RoboRacer (F1TENTH) standard, plus a cardboard lab track: **\$1,865 recommended**, \$1,300 minimum, \$3,400 for the official reference build. The FSAE car is on a two-year cycle and is not available for autonomous work this season; this car is what the autonomous team develops, tests and learns on in the meantime, and the pipeline built on it is the one that goes onto the FSAE car next year.

1 What the car is made of

Part	What it does
Chassis	Traxxas Fiesta ST Rally VXL, ready to run: the Slash 4x4 low-CG chassis under a rally body, with 4WD drivetrain, steering servo, brushless motor and radio. The chassis the current RoboRacer build guide and deck drawings are made for, and the car our simulator models.
Motor controller	A VESC-type controller replaces the stock speed controller. Drives the motor and the steering servo from USB commands and reports motor rotation back, which is the car's speed. The part that turns a toy into a robot.
Computer	NVIDIA Jetson Orin Nano with an NVMe drive. Runs ROS 2 and all software: sensor drivers, perception, planning, control.
2D lidar	Spins a laser and returns a range per angle, 10 to 40 times a second. Sees walls and cones; enough to drive with no map.
Camera	Colour and texture the lidar lacks: blue cone or yellow cone. Needed for the cone-track perception work.
Joystick	A Bluetooth gamepad on the computer for manual driving and the dead-man switch: the car only drives autonomously while a button is held. This is how the RoboRacer software is set up; the chassis's own radio stays in the box.
Power	One lithium pack drives the motor through the controller; the Jetson takes the pack voltage directly through a fuse, and a 5 V step-down converter feeds the lidar and camera. A switch on that branch and a safety bag for charging.
Deck	Laser-cut plates and standoffs from the published drawings; every part has a place.
Track	Cardboard walls at lidar height, taped into loops; runs in any corridor. Cones are added for the camera work.

Sensor to actuator

1. **Sense.** Lidar driver publishes a scan; VESC driver publishes wheel odometry and IMU; camera driver publishes an image.

2. **Perceive.** Walls or cones located relative to the car; cones classified by colour from the image and ranged by the lidar; later, position on a map.
3. **Plan.** A target bearing and speed: the widest gap at first, a point on a raceline later.
4. **Control.** Steering angle from the bearing, throttle from the speed error against odometry.
5. **Actuate.** The VESC drives the servo and the motor, gated by the joystick dead-man. Every topic is recorded for replay.

The nodes and topic names are the same as in the team's simulator, so a stack developed there is dropped onto the car with only the driver nodes swapped.

2 Options and cost

Three grades of the same car, following the official RoboRacer bill of materials with cheaper sensor and controller choices where they do not change what the car can teach. Every line is one purchase; the link is the maker's page.

Item	Minimum	Recommended	Reference
Chassis: Traxxas Fiesta ST Rally VXL , ready to run	430	430	430
Controller: Flipsky Mini FSESC 6.7 / VESC 6 MkVI	110	110	250
Compute: Raspberry Pi 5 8 GB kit / Jetson Orin Nano Super + 500 GB NVMe	130	350	350
Lidar: RPLidar A1M8 / RPLidar S2 / Hokuyo UST-10LX	100	400	1,200
Camera: Logitech C920 / Luxonis OAK-D Lite / RealSense D435i	60	150	335
Power: two Zeee 3S 6000 mAh packs, balance charger, LiPo bag	135	135	135
Power distribution: 5 V converter, barrel-jack pigtail, balance extender, fuse, switch / RoboRacer powerboard	40	40	125
Joystick (8BitDo or DualShock), antenna, PPM cable, bullet adapters, USB cables, display emulator	70	70	70
Deck plates (cut on campus from the RoboRacer drawings), standoffs, M3/M5/M2.5 fastener kits	90	90	90
Track: cardboard walls, floor tape	15	30	30
Spares: tyres, pinion, bumper	40	60	60
Total	1,220	1,865	3,075

Minimum proves the pipeline: lidar-only driving on a Pi at about 2 m/s. **Recommended** is the official build with a time-of-flight RPLidar and a Flipsky controller in place of the two expensive parts; every item is one the team keeps. **Reference** is the official RoboRacer competition bill of materials with a camera added; the Hokuyo lidar is most of the difference and is not needed to learn on.

The lidar choice

The lidar sets the speed the car can drive safely: at 10 to 12 Hz the car moves about 25 cm between scans at 3 m/s, which is the practical limit for the cheaper units.

Lidar	USD	Rate	Range	Notes
RPLidar A1M8	100	5 to 10 Hz	12 m	proves the pipeline, nothing more
RPLidar A2M12	320	12 Hz	12 m	triangulation; fine indoors, noisy on dark surfaces
RPLidar S2	400	10 Hz	30 m	time of flight; robust to dark surfaces and daylight
RPLidar A3	600	15 to 20 Hz	25 m	the fastest of the cheap units
Hokuyo UST-10LX	1,200	40 Hz	10 m	the reference; industrial, lasts indefinitely

All publish the same ROS message, so swapping lidars later changes no code.

3 Why it is worth it

For the autonomous team this semester

- The FSAE car is being built and is not available for autonomous work; without a platform the team has nothing to run on until next year.
- Members stay for what they can see move. A lap time keeps someone a semester; a car that turns on their code keeps them for years.
- Real dynamics and real sensors cannot be simulated: grip that changes with the floor, servo backlash, battery sag, a lidar that sees chair legs and reflections.
- New members learn wiring, power, safety, motor control and ROS 2 on hardware at low cost and low risk.
- The simulator and the car share one codebase, so every member's stack has a place to go.

Carrying over to the FSAE car next year

- **Same pipeline.** Cone detection with camera plus lidar, a midline between cones, a raceline, a tracking controller: the Formula Student Driverless stack, at small scale.
- **Same tools.** ROS 2 nodes and topics, sensor drivers, bag recording and replay, sim-to-real testing, safety procedures. These transfer directly.
- **Same method.** Develop in the simulator, transfer to the car, measure the gap, close it. The team arrives at the FSAE car already practised at this.
- **What does not transfer:** vehicle dynamics and tuned parameters. A 3 kg car says nothing about a 250 kg car's tyres, and the proposal does not claim it does.

4 Will it work

The build is the F1TENTH/RoboRacer platform, assembled by hundreds of university teams since 2016 from a public build guide, drawings and drivers; the parts list above is that guide's own bill of materials with cheaper sensor and controller choices. Driving laps of a lab track is a solved problem, and the team's simulator already runs the same car, sensors and software. The risks are elsewhere: the car needs one named owner for assembly, batteries and access, or it goes unused; integration takes two to four weekends before the first autonomous lap; laps come quickly but mapping, racelines and perception are real work and will fill the semester; crashes break bumpers and, occasionally, the lidar, which the spares line covers. With an owner, the chance of the car never driving is low; the more likely outcome is a car that laps within a month and a team spending the rest of the semester making it faster and smarter.

5 Options set aside

- **The team's Tamiya chassis.** A motorless touring-car kit. Saves about \$300 net for a semester of mechanical work with no drawings to copy, does not match the simulator's car, and is not eligible for any RoboRacer event. Kept as a bench mule.
- **A 1:4 or 1:5 car.** No 1:4 road car exists; the nearest are 1:5-class trucks (Traxxas X-Maxx, Arrma Kraton 8S, Losi 5IVE-T). Closer to FSAE in size and able to carry the team's RoboSense M1 lidar, but about \$3,500 even with the M1 and the Alameda test track counted, and a 10 kg car at 50 km/h cannot be run on campus: every test is an off-site session. A credible second step once the 1:10 programme is running, not a first car.

RoboRacer build guide and bill of materials: <https://f1tenth.readthedocs.io>

Team simulator and manuals: <https://github.com/Pranman1/feb-racing>